César Garin

EPFL — Lausanne, Switzerland | Swiss residence permit

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SUMMARY

EPFL Computer Science and Communication Systems student with a strong quantitative foundation in probability, statistics, machine learning, algorithms, and scientific computing. Builds independent projects across predictive modelling, NLP, reinforcement learning, and financial data analysis. Currently on a one-year academic exchange at DTU in Copenhagen, focusing on statistical learning, data science, and quantitative modelling. **Available for a summer internship from June 2027.**

WORK EXPERIENCE

Independent Entrepreneur — **Lyon, France**
Event Services — *Jun 2026 – Present*

- Independently operate a **service business** providing services for **weddings, professional events and private events.**
- Generated **€4,000+** in revenue since launching the activity.

Forum EPFL — **Lausanne, Switzerland**
Logistics Staff — *Oct 2025*

- Coordinated operations for **150+ partner companies** participating in Switzerland's largest student recruitment forum.
- Worked in a high-volume professional environment with representatives from **banking, consulting, technology.**

PolySI EPFL — **Lausanne, Switzerland**
Supervision Staff — *Aug 2025*

- Led and supervised groups of incoming students during EPFL's integration week. Managed activities, logistics, and unexpected situations involving large groups of students.

Getronics — **Lyon, France**
Intern — *Jan 2024 – Mar 2024*

- Completed **cross-departmental** rotations within an international IT services company.
- Gained exposure to corporate finance, revenue mechanisms, budgeting, cost analysis, and procurement..

EDUCATION

EPFL — École Polytechnique Fédérale de Lausanne — **Lausanne, Switzerland**
B.Sc. Computer Science & Communication Systems — *Sep 2024 – Jun 2027*

- Coursework: Machine Learning, Algorithms, Probability & Statistics, Computer Architecture, Software Construction, Computer Systems **(TCP/IP networking)**, Linear Algebra and Mathematical Analysis.

DTU — Technical University of Denmark — **Copenhagen, Denmark**
B.Sc. Computer Science – Academic Exchange Year — *Sep 2026 – Jun 2027*

- Coursework: Machine Learning for Signal Processing, Algebra, Intelligent Systems, Digital Communication, Database Systems, Logic for Security, Algorithms for Massive Data Sets, Software Development.

SELECTED PROJECT

ML × Quantitative Finance Research Series — **2026**
Personal Research

- Developed a **four-project research series** applying machine learning to financial modelling, covering **XGBoost** equity price prediction, **NLP** sentiment analysis with **FinBERT**, **PPO reinforcement learning** for derivative hedging, and **neural networks** for implied volatility surface modelling.

NanoGPT — Transformer Language Model from Scratch — **2026**
Personal Project

- Implemented a **GPT-style transformer architecture** from scratch in **PyTorch**, including **multi-head self-attention, residual connections, layer normalization, and feed-forward blocks.** Built and trained the model of approximately **10 million parameters** on the Tiny Shakespeare dataset.

Mini RISC-V Processor — **Sep 2025 – Jan 2026**
EPFL Project

- Designed and simulated a **CPU pipeline** in Verilog as part of EPFL coursework. Implemented instruction execution, memory management, and interrupts using **RISC-V Assembly.**

SKILLS & LANGUAGES

- **LLM Engineering: RAG** & vector search, agent orchestration & tool calling, prompt design, LLM evaluation.
- **Operating Systems:** Linux, Windows, macOS.
- **Programming: Python**, C, Java, Scala, R, LaTeX.
- **Machine Learning: PyTorch**, scikit-learn, **XGBoost**, FinBERT, Reinforcement Learning (**PPO**).
- **Data & Scientific Computing:** NumPy, Pandas, MATLAB, Jupyter Notebook.
- **Tools: Git**, Excel, PowerPoint, Overleaf.
- **Languages:** French (native), English (**C2 – CEFR**), German (**Intermediate**). **Interests:** Competitive Skiing, Football, Weightlifting & Martial Arts, Electric guitar & Piano.